



RESEARCH KNOWLEDGE REPOSITORY WITH SEMANTIC SEARCH

GROUP-4

SUPERVISOR:
MRS.MELINDA

25CS1302E
DBE & DSD

TEAM MEMBERS
2510080024-AKSHAYA
2510080022-NIHARIKA
2510080019-HARSHINI
2510080023-VARNA

PROBLEM STATEMENT

- Traditional search systems rely on exact keyword matching, which often misses relevant research papers.
- Different words or synonyms for the same concept can prevent important papers from appearing in search results.
- The large number of published research papers makes finding relevant information difficult and time-consuming.
- Keyword-based searches often return many irrelevant papers, reducing search efficiency.
- Traditional search engines do not understand the context or meaning of user queries, leading to less accurate results.

Target users:

- Students
- Researchers
- Faculty members
- Research organizations
- Digital libraries

Importance:

- Improved marketing strategies
- Better inventory management
- Increased sales performance

NEED FOR THE PROJECT

(SOCIAL NEED)

Why is this project needed?

- Makes research faster and more accurate.
- Reduces time spent searching for relevant papers.
- Improves access to knowledge.

Beneficiaries

- Universities , Colleges
- Libraries
- Researchers
- Companies with internal knowledge bases

Impacts

- Better research productivity.
- Improved knowledge sharing.
- Supports innovation and informed decision-making.

PROJECT OBJECTIVES

1

To Develop a centralized repository for research documents.

2

To Enable uploading and management of research papers.

3

To Implement semantic search to understand user intent

4

To Rank documents based on relevance.

5

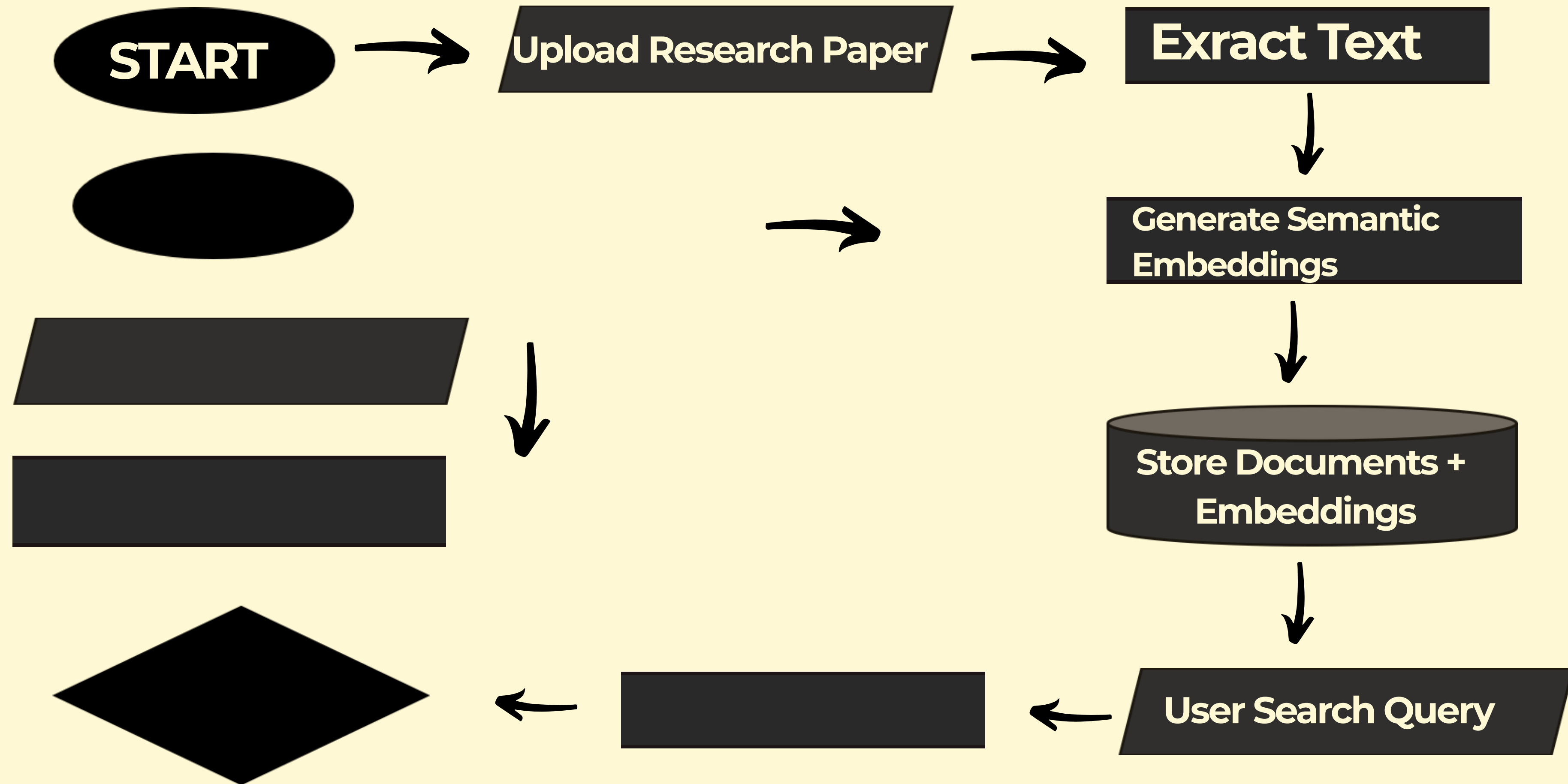
To Provide an easy-to-use interface for searching and accessing research papers

PROPOSED SOLUTION / METHODOLOGY

The system stores research documents and generates semantic embeddings for each document. When a user searches, the system compares the query embedding with stored document embeddings to retrieve the most relevant results based on meaning rather than exact keywords.

MAIN MODULE

- **User Authentication:** Verifies user identity and provides secure access to the system.
- **Document Upload:** Allows users to upload research papers for processing and storage.
- **Text Extraction:** Extracts readable text content from uploaded research documents.
- **Embedding Generation:** Converts extracted text into semantic vector embeddings for efficient searching.
- **Semantic Search:** Finds research papers based on meaning and context rather than exact keywords.
- **Search Result Ranking:** Ranks retrieved papers according to their relevance to the user's query.
- **Repository Management:** Organizes, stores, and manages research papers and their associated embeddings.



FEASIBILITY ANALYSIS

IDE
VS Code

Framework
FastAPI/Flask
React

Hardware

- Laptop/Desktop
Windows 11 OS
intel core 5
- Connection
- Minimum 8 GB
RAM

Software

- Programming
Language
- Python version
3.14
 - JavaScript
ES2026

Database
PostgreSQL or MongoDB

Libraries
Sentence Transformers
FAISS
PyPDF2
NumPy

Dataset

Initially, use:

Research papers downloaded from publicly available sources
(such as arXiv or open-access repositories), or
A self-created collection of PDFs for demonstration.

NOVELTY OF THE PROJECT

- Unique Features :
- Searches based on meaning instead of exact keywords.

Retrieves relevant documents even when different terminology is used.

Uses AI-powered semantic embeddings.

Provides more accurate ranking than traditional keyword search.

Can be extended with AI features such as document summarization, question answering, and recommendations.

Difference from Existing Systems :	Traditional Search	Proposed System
	Keyword matching	Semantic understanding
	Lower relevance	Context-aware search
	Limited context	Higher relevance
	Exact word dependency	Meaning-based retrieval

THANK YOUU !!!